



Ultra-wide bandwidth system optical fiber channel modeling based on Kolmogorov-Arnold network and physical constraints

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Abstract: The rapid growth of artificial intelligence (AI) has placed stringent demands on fiber-optic backbone networks supporting data transmission for intelligent computing centers. In optical fiber system design, achieving channel simulations that are both highly accurate and efficient is crucial. Building on the separable operator network framework, we introduce the Kolmogorov–Arnold Network (KAN), which outperforms traditional multi-layer perceptron (MLP) in modeling physical equations. Leveraging this advantage, we construct what we believe to be a novel separable operator KAN (SepOKAN) architecture. Furthermore, by incorporating physical constraints into SepOKAN, we develop a separable physics-informed KAN (SPIKAN) architecture that offers greater interpretability and improved simulation accuracy. We validate the proposed models in a stepwise manner: first in a single-channel dual-polarization scenario for fundamental verification, and then in an ultra-wideband (UWB) multi-channel scenario. We evaluate fitting accuracy and generalization performance through multi-dimensional comparisons. Finally, experiments on real-world single-channel and wavelength-division multiplexing (WDM) fiber link systems demonstrate that the SPIKAN model achieves high accuracy and robust performance, highlighting its practical value for optical fiber communication system design.

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1. Introduction

Optical fiber communication networks are evolving rapidly to accommodate the explosive growth of data traffic driven by AI and next-generation technologies. Modern fiber-optic backbone networks are expected to serve massive users with ultra-high speed and ultra-low latency [1] [2]. The fiber link, as the core transmission medium, plays a pivotal role in network performance. Consequently, accurate and efficient fiber channel modeling has become paramount for developing new algorithms, understanding signal impairments, and reducing experimental costs.

Currently, the split-step Fourier method (SSFM) [3] is the dominant numerical technique for solving the nonlinear Schrödinger equation (NLSE) and simulating optical signal propagation. SSFM iteratively computes fiber attenuation, dispersion, and nonlinearity along the link to achieve high modeling accuracy [4]. However, this iterative procedure must handle numerous fiber parameters and complex nonlinear interactions, leading to substantial computational overhead and increased complexity [5]. In pursuit of faster simulations, analytical models like the Gaussian noise (GN) [6] and enhanced GN (EGN) [7] have been proposed. These simplify performance evaluation via mathematical approximations [8]. Yet, such models cannot fully capture transient

waveform details and remain constrained by theoretical assumptions [9]. This gap indicates considerable room to improve generality and accuracy across diverse fiber scenarios.

In recent years, the integration of AI with optical communications has spurred new modeling approaches [10]. Fiber channel modeling strategies can be broadly categorized by their use of data or physics. (1) Data-driven methods: These rely purely on channel input-output data to train neural networks, achieving high precision without explicit physical models [11]. For example, bi-directional long short-term memory (BiLSTM) [12], generative adversarial network (GAN) [13], and Transformer-based [14] models have been used to approximate fiber channel responses. However, data-driven models demand large labeled datasets and lack physical interpretability. (2) Physics-driven methods: These incorporate the governing physics into model training via physics-informed neural network (PINN). By using known initial/boundary conditions and embedding physical laws, physics-driven models can generalize well even with limited data [15]. For instance, a PINN-based approach in used time and distance as inputs under NLSE guidance to achieve a highly generalized fiber model [16]. The trade-off is that such models often require more training iterations but maintain physical consistency in data-sparse regimes. (3) Hybrid physics-data methods: These combine data-driven flexibility with physical robustness. A prominent example is operator learning, which learns mappings between function spaces [17]. Techniques like the fourier neural operator (FNO) have simulated long-distance fiber transmission of high-order QAM signals [18], and deep operator network (DeepONet)'s two-branch structure has achieved waveform-level accuracy for single-channel and C+L band WDM systems [19]. To improve efficiency, a separable operator network (SepONet) was introduced, exhibiting higher computational efficiency than DeepONet for complex partial differential equation (PDE) problems [20]. Despite this progress, most neural-network-based fiber models still use conventional MLP architectures. The recently proposed KAN offers a new paradigm: it replaces fixed activation functions with learnable ones and parameterizes weights as spline functions [21]. Studies show that a compact KAN can achieve accuracy comparable to or better than a much larger MLP, especially on PDE-related tasks, while also improving computational efficiency [22]. This suggests that applying KAN to fiber channel modeling could significantly reduce computational load and training time without sacrificing accuracy. KAN has demonstrated superior performance in nonlinear compensation [23], nonlinear waveguides [24], and optical systems [25] [26].

In this work, we redesign the neural network modeling approach by integrating KAN into the fiber-channel SepONet framework and further infusing physical knowledge. First, we adapt the SepONet operator-learning architecture to the fiber channel modeling task. Next, we incorporate the KAN's learnable activation mechanism into SepONet, yielding a hybrid SepOKAN architecture. Finally, we introduce a physics informed loss term to build SPIKAN, a SepOKAN variant guided by physical principles. The SPIKAN model enhances interpretability and accuracy by enforcing the NLSE-based constraints during training. We validate our models progressively: Section 3 examines a long-haul single-channel system, demonstrating training convergence and accuracy improvements compared to existing models. Section 4 then extends the approach to an UWB system, where we develop a physical constraint framework to account for inter-channel Raman scattering and evaluate performance across channels. In Section 5, we test SPIKAN on real single-channel and WDM fiber link experiments to verify its practical effectiveness. Our results show that SPIKAN achieves excellent agreement with SSFM in both time and frequency domains, with significantly lower computational complexity than SSFM and other neural networks. It maintains low normalized mean square error (NMSE) over long distances and across varying power levels and modulation formats, highlighting the model's robustness. Section VI concludes the paper by summarizing contributions and discussing future research directions.

2. Principle

In this chapter, we briefly review the fundamental theories and traditional methods of optical fiber modeling and provide a detailed introduction to the designed neural network architecture.

2.1. Optical fiber transmission model

In long-haul optical fiber transmission systems, the impact of polarization mode dispersion (PMD) accumulates progressively with increasing transmission distance, making it a critical factor that cannot be overlooked in system modeling. Furthermore, under large-bandwidth transmission conditions, the stimulated Raman scattering (SRS) effect becomes significant due to the expanded bandwidth. Therefore, this paper takes into account both PMD and SRS effects. The transmission of single-channel dual-polarization optical signals in fiber links can be described by the Manakov equation, expressed as follows [27]:

$$\frac{\partial E(z, t)}{\partial z} = -\frac{\alpha}{2}E(z, t) - j\frac{\beta_2}{2}\frac{\partial^2 E(z, t)}{\partial t^2} + j\frac{\beta_3}{6}\frac{\partial^3 E(z, t)}{\partial t^3} + j\gamma\frac{8}{9}|E(z, t)|^2E(z, t) - \frac{\Delta\beta_1}{2}\sigma\frac{\partial E(z, t)}{\partial t}, \quad (1)$$

where $E(z, t)$ represents the optical field over two arbitrary orthogonal polarization modes $E_x(z, t)$ and $E_y(z, t)$, α denotes the attenuation coefficient of the fiber channel, β_2 signifies the dispersion coefficient of the fiber, β_3 signifies the third-order dispersion, γ indicates the nonlinear coefficient of the fiber, $\Delta\beta_1$ stands for the differential group delay (DGD), and σ describes the coupling between the two polarization states, where $\sigma = [\sigma_{xx}, \sigma_{xy}; \sigma_{yx}, \sigma_{yy}]$ is a unitary matrix.

Since Eq. (1) does not have an analytical solution, the SSFM is generally employed for numerical solving. The expression for the symmetric SSFM with a step size of h is as follows [28]:

$$\begin{cases} E(z+h, t) = E(z, t) \exp\left(\frac{h}{2}\hat{D} + \int_z^{z+h} \hat{N}(\epsilon)d\epsilon + \frac{h}{2}\hat{D}\right), \\ \hat{D} = -\frac{\alpha}{2} - j\frac{\beta_2}{2}\frac{\partial^2}{\partial t^2} - \frac{\Delta\beta_1}{2}\sigma\frac{\partial}{\partial t}, \\ \hat{N} = j\gamma\frac{8}{9}|E(z, t)|^2. \end{cases} \quad (2)$$

In UWB transmission systems, a closed-form analytical solution for SRS was proposed, initially validated in the $C + L$ band [29], and later further confirmed in $S + C + L$ UWB transmission experiments [30]:

$$\frac{\partial P_i}{\partial z} = -\sum_{b=a+1}^k \frac{f_b}{f_a} g_r(\Delta f) P_b P_a + \sum_{b=1}^{a-1} g_r(\Delta f) P_b P_a - \alpha(f_i) P_a, \quad (3)$$

where $g_r(\Delta f)$ is the Raman gain spectrum, $\Delta f = |f_a - f_b|$ is the frequency separation, the P_i is the signal power of different channels, $\alpha(f_i)$ represents the frequency-dependent fiber attenuation.

2.2. Principle of neural network

The optical fiber transmission process can be fundamentally described as a mathematical physical process governed by physical equations, where input signals are mapped to output signals through specific mathematical operators. The Kolmogorov-Arnold theorem states that any multivariate continuous function can be expressed as a composition of univariate functions. Leveraging this principle, KAN employ learnable activation functions to achieve such functional decomposition, thereby exhibiting more efficient representational capabilities and enhanced physical consistency in tasks such as PDE modeling. As depicted in Fig. 1, KAN exhibits fundamental architectural distinctions from MLP. The MLP is constructed by nesting basic units shown in Fig. 1(a). Given

a n -dimensional input vector $X = [x_1, x_2, x_3, \dots, x_n]$, the output of the k -th layer with weight matrix $\mathbf{W}^{(k)}$, activation function $\sigma(\cdot)$, and bias vector $\mathbf{b}^{(k)}$ is expressed as:

$$\begin{cases} Z^1 = \sigma(W^1 X + b^1), k = 1 \\ Z^k = \sigma(W^k Z^{(k-1)} + b^k), k > 1 \end{cases} \quad (4)$$

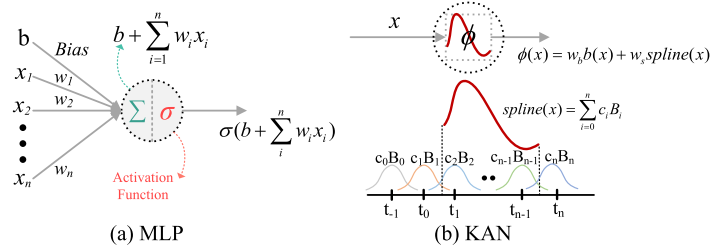


Fig. 1. The foundational architectures of the MLP and the KAN. (a) The basic structure of the MLP, (b) The fundamental architecture of the KAN.

As illustrated in Fig. 1(b), any continuous multivariate function defined on a bounded domain can be represented as a combination of a finite number of univariate functions and a set of linear operations. For any smooth function $f : [0, 1]^n \rightarrow \mathbb{R}$, there exist continuous univariate functions ϕ_q and $\Phi_{q,p}$ such that [31]:

$$f(x) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right), \quad (5)$$

where $\phi_{q,p} : [0, 1] \rightarrow \mathbb{R}$, $\Phi_q : \mathbb{R} \rightarrow \mathbb{R}$. In the details of the KAN implementation, as shown in Fig. 1(b), the activation function consists of two parts: the base function $b(x)$ and the linear combination of spline function $spline(x)$:

$$\begin{cases} \phi(x) = w_b b(x) + w_s spline(x), \\ b(x) = silu(x) = \frac{x}{1 + \exp^{-x}}, \\ spline(x) = \sum_i c_i B_i(x), \end{cases} \quad (6)$$

where w_b and w_s are the weight, c_i are the trainable parameter, $B_i(x)$ are the spline function, $silu$ is the Sigmoid Linear Unit function. The order of the spline function and the selected combination grid interval numbers are important factors affecting the complexity of the KAN network. The construction method of the KAN network has been detailed in [21], and all KAN in this paper is implemented following the methodology described therein.

To learn the operator of optical fiber channel transmission, we use the SepONet architecture. SepONet typically consists of one branch network and two trunk networks. SepONet achieves operator learning by separately learning basis functions for different coordinate axes. As the structure and principles of SepONet have been thoroughly detailed in [20], we will not elaborate further here, but it is worth noting that our SepONet includes four branch networks. As the complex-valued signal needs to be decoupled into real and imaginary components for both X- and Y-polarization states, resulting in four independent real-valued equations, four dedicated networks are accordingly designed and assigned to learn these equations respectively. The specific implementation details will be elaborated in subsequent sections. The rest of the principles remain consistent with the original version.

2.3. Neural network channel modeling architecture

To model the fiber channel, we adopt a two-stage approach. First, we design a sliding window strategy to construct training samples from the simulated transmission data, as shown in Fig. 2(b). Specifically, we take the complex dual-polarization signal and split it into real and imaginary parts, yielding four real-valued data streams. Within each sliding window, we place the X-polarization signal and its corresponding Y-polarization signal side by side, thereby creating input-output pairs that capture the coupling between polarizations. This sliding window moves through the data, the window length set to 16 symbols, the stride set to 4 symbols in the SepONet configuration, generating a rich dataset of paired sequences. Next, we develop a neural network architecture informed by optical physics and train it on the constructed dataset. As shown in Fig. 2(c), the architectural schematic of a single Trunk network and Branch network is illustrated. The network is based on the separable operator learning concept of SepONet but extended to two trunk networks and four branch networks, with each branch learning one of the four decoupled real-valued equations derived from the Manakov model. The overall architecture is illustrated in Fig. 2(d). As shown in Fig. 2(d), the BN is the branch net, the physical coordinates z and t are first input into two separate Trunk networks to generate intermediate outputs $\tau_{z,k}^{(\cdot)}$ and $\tau_{t,k}^{(\cdot)}$, whose Kronecker product $\tau_{z,k}^{(\cdot)} \otimes \tau_{t,k}^{(\cdot)}$ is computed. Concurrently, the four real-valued components of the system-exported dual-polarization complex-valued signal are fed into four Branch networks, producing outputs $\beta_k^1, \beta_k^2, \beta_k^3, \beta_k^4$. The final network output is derived by performing element-wise multiplication between $\tau_{z,k}^{(\cdot)} \otimes \tau_{t,k}^{(\cdot)}$ and each β_k , followed by summation across all branches. The number of transmitted symbols is set to 2^{13} .

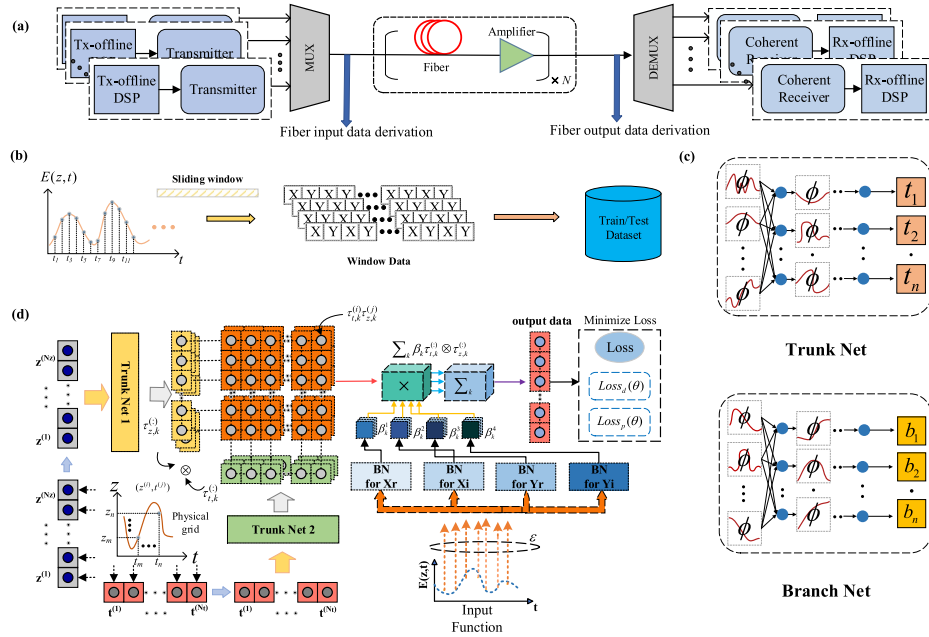


Fig. 2. SPIKAN fiber channel modeling. (a) S + C + L-band system, (b) Dataset construction, (c) Trunk and Branch network architectures, (d) SPIKAN modeling scheme.

This physics-guided design ensures that each branch specializes in a specific subdynamics of the fiber system, aligning the model structure with the underlying equations. We initially configure the SepONet model with five hidden layers of 256 neurons each, using the Adam optimizer, the learning rate set to 0.001 and mean squared error (MSE) loss. For data efficiency,

each training batch contains 256 samples. Building upon this baseline, we construct SepOKAN by introducing the KAN activation mechanism into the network. SepOKAN uses a more compact architecture: three hidden layers of 64 neurons each, and it replaces the fixed SiLU activations with third-order B-spline activations with a grid interval of 6. To better balance model accuracy and computational complexity, we optimized the data processing of SepOKAN by reducing the sliding window length to 8 symbols while maintaining a stride of 4 symbols, and decreasing the batch size to 128. Finally, to enhance physical consistency, we incorporate a physics-based loss alongside the data-driven loss. The total loss is defined as a weighted sum of the data fitting loss L_d and the physics-informed loss L_p :

$$Loss = w_d Loss_d + w_p Loss_p, \quad (7)$$

where w_d and w_p balance the two terms, represent the weights for the pure data constraint and the physical constraint, respectively. In our implementation, w_d and w_p are adjusted adaptively during training, the method described in [32]. This SPIKAN architecture ensures that the learned model not only fits the data but also obeys known physical laws of fiber propagation. For multi-span fiber links, we employ a cascaded modeling approach: the output of each span's network serves as the input to the next span, thereby accurately simulating the accumulated effects over distance. All network weights are initialized via Kaiming initialization and biases set to zero to facilitate stable convergence [33].

We evaluate model performance primarily using the NMSE metric, defined as [19]:

$$NMSE = \frac{\|\hat{s} - s\|_2^2}{\|s\|_2^2}, \quad (8)$$

where $\hat{s}$ and s denote the predicted and true signal vectors, both $\hat{s}$ and s represent waveform sampling points. An NMSE closer to 0 indicates better accuracy. Corresponding to the single-channel and UWB transmission scenarios, this study designs dedicated physics-based constraint loss functions respectively, whose specific mathematical expressions will be elaborated in Section 3 and Section 4. With this setup, the following sections will validate the proposed SPIKAN model against baseline methods in various scenarios.

3. Long-haul single-channel system

First, the effectiveness of the proposed scheme is validated on a single-channel dual-polarization system. For this scenario, a systematic comparative analysis is conducted against various neural network models, encompassing multiple dimensions including training convergence, fitting accuracy, and generalization capability.

3.1. Single-channel dual polarization parameter setting

A single-channel dual-polarization system is constructed, and simulation parameters are configured, with its parameters configured according to Table 1, where n_2 is nonlinear refractive index. Unless otherwise specified in the following sections, the data at 0 dBm is used for training purposes throughout the remainder of this work. The dataset is extracted following the method illustrated in Fig. 2. The launch power is set to 0 dBm. Since the single-channel scenario does not involve energy transfer between different wavelength channels, SRS effects are not considered in the simulations of this section. The optical carrier frequency is set at 1550 nm for 15 spans signal transmission. To prevent the neural network from recognizing periodic patterns in pseudorandom sequences, the Mersenne Twister algorithm is employed in this study to generate the pseudorandom sequences [34]. This algorithm is characterized by its high randomness and long periodicity, effectively mitigating the interference of pseudo-random sequence patterns on signal modeling. The four branch networks designed in this paper are used to learn the four

operators from Eq. (16) in the Appendix. To accelerate network convergence and enhance the physical interpretability of the network, this paper designs the SPIKAN for the characteristics of single-channel dual-polarization systems and constructs physical constraints. Therefore, the loss function of SPIKAN in single-channel systems is:

$$loss_d = \frac{1}{N_d} \sum_{i=1}^{N_d} |\hat{s}_i - s_i|^2, loss_p = \frac{1}{N_p} \sum_{i=1}^{N_p} (|f_1|^2 + |f_2|^2 + |f_3|^2 + |f_4|^2), \quad (9)$$

where f_1 to f_4 correspond to Eqs. (22) to (23) in the Appendix. The neural network's output format matches the fiber terminal configuration, enabling direct computation of the data-driven loss component $loss_d$. Using the derived f_1, f_2, f_3 and f_4 from Appendix A and the fiber parameters listed in Table 1, the physics-informed loss $loss_p$ is calculated by substituting into Eq. (9). This $loss_p$ value is then substituted into Eq. (7) to compute the total loss function of the neural network.

Table 1. System Partial Simulation Parameters

Parameter	Value	Unit	Parameter	Value	Unit
Span Length	80	km	Dispersion	17	ps/nm/km
Symbol Rate	30	GBaud	Noise Figure	0	dB
Core Area	80×10^{-12}	m ²	OSNR	30	dB
Attenuation	0.22	dB/km	Dispersion Slope	0.08	ps/(nm) ² /km
n_2	2.6×10^{-8}	μm ² /W	Raman fraction	0.18	-

3.2. Model accuracy

This study systematically evaluates the training performance of five neural network models for optical fiber channel modeling: GAN [12], DeepONet [19], SepONet, SepOKAN, and SPIKAN. To ensure modeling accuracy, structural adjustments were made to DeepONet and GAN: DeepONet was configured with an input dimension of 256 sampling points and a 5-hidden-layer architecture, each with 512 neurons; GAN adopted a 9-layer network structure. The training results indicate that SepONet, SepOKAN, and SPIKAN exhibit rapid initial decline in NMSE, yet their test curves show significant fluctuations, gradually stabilizing after approximately 20,000 training epochs, as shown in Fig. 3. The GAN approach achieves the fastest convergence speed, reaching a stable state within a relatively short duration. DeepONet demonstrates the most stable training characteristics, achieving convergence around 15,000 epochs with highly consistent training and testing curves throughout. The test curves of SepONet, SepOKAN, and SPIKAN exhibit some fluctuations during the initial training phase but eventually converge to a low and stable NMSE level after sufficient iterations. In terms of convergence performance, once stability is achieved, all three networks demonstrate progressively improving modeling capabilities. Among the five compared methods, SPIKAN demonstrates the most outstanding performance, achieving the lowest NMSE value upon convergence. This superior performance primarily stems from the incorporation of physical constraints, which enables it to attain a better solution than SeqOKAN.

When the NMSE reached 6.01×10^{-4} , a comparative analysis is conducted between the outputs of SPIKAN and SSFM in terms of time domain, frequency domain, and demodulation performance, as illustrated in Fig. 4. The results demonstrate that the simulated outputs of SPIKAN in both frequency and time domains closely align with the SSFM benchmark results, with only minimal discrepancies observed in the constellation diagrams. The reconstructed constellations after digital signal processing show virtually no additional distortion, further demonstrating the accuracy of the proposed modeling method. After digital signal processing

(DSP), the BER values were measured as 2.26×10^{-2} for SSFM and 2.36×10^{-2} for SPIKAN, indicating comparable performance and validating the high accuracy of the proposed modeling approach.

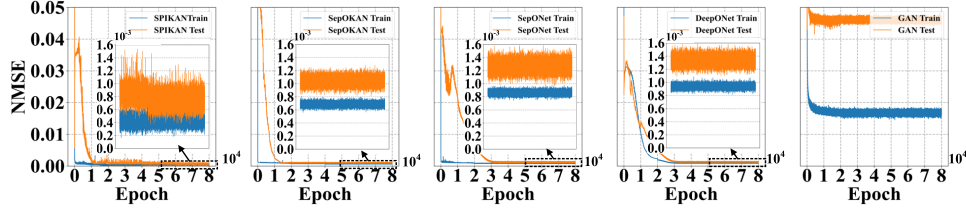


Fig. 3. Changes of NMSE of five different neural networks with epochs.

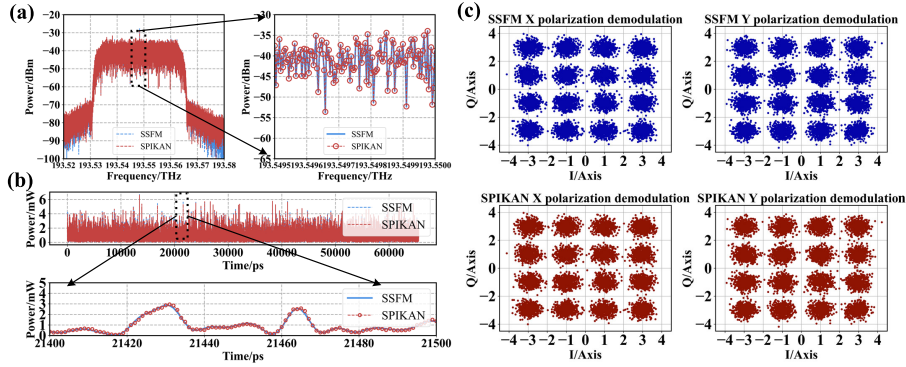


Fig. 4. Comparison of results. (a) Frequency domain SSFM and SPIKAN, (b) Time domain SSFM and SPIKAN, (c) Constellation diagram SSFM and SPIKAN.

3.3. Generalization ability

The primary objective of neural networks is to learn the mapping relationship between inputs and outputs, enabling them to generate reasonable outputs based on the learned mappings even when input conditions change. Here, we focus on studying the impact of factors such as transmission power, modulation format, and amplified spontaneous emission (ASE) noise on the model's generalization ability. In Fig. 5, we employ data ranging from -5 dBm to 5 dBm for generalization using the model trained under 0 dBm condition.

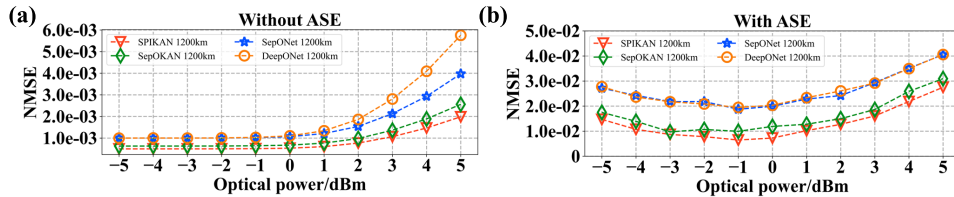


Fig. 5. The ability to generalize signals. (a) without consideration of ASE, (b) with consideration of ASE.

The generalization performance of DeepONet, SepONet, SepOKAN, and SPIKAN was evaluated on a dataset with a noise figure of 0 dB. The results indicate that all models maintained relatively stable NMSE in the linear region. However, as the power entered the nonlinear regime,

fiber nonlinear effects led to increased signal distortion, resulting in degraded generalization capability and elevated NMSE values. Notably, the modeling performance of DeepONet and SepONet deteriorates rapidly, while even SepOKAN, which demonstrates relatively better performance, shows inferior results compared to SPIKAN. SPIKAN sustained an NMSE on the order of 1×10^{-3} even under a 1200 km long-haul transmission scenario, as illustrated in Fig. 5(a). In practical optical transmission systems, amplifier noise introduced by optical amplifiers cannot be neglected. We evaluate the generalization capability of the proposed networks using a dataset with a noise figure of 5 dB. In the linear region, signal quality is primarily affected by ASE noise; as power increases to the optimal transmission level, signal quality improves accordingly, as illustrated in Fig. 5(b). However, when power further enters the nonlinear regime, nonlinear effects become dominant and interact with ASE noise, leading to significant signal degradation. However, SPIKAN consistently maintains optimal performance, achieving the lowest NMSE throughout the entire power range and demonstrating remarkable noise and nonlinear robustness.

Optical communication systems typically involve multiple modulation formats, necessitating models with format-agnostic generalization ability. The model was trained on 16QAM data and tested on 4QAM, 32QAM, and 64QAM data to assess prediction accuracy. Given the close relationship between fiber channel characteristics and overall system design, the Q-factor is introduced herein as a system-level performance evaluation metric. The quantitative relationship between the Q-factor and the bit error rate (BER) can be expressed as:

$$Q = 20\log_{10}(\sqrt{2}\text{erfc}^{-1}(2\text{BER})), \quad (10)$$

where erfc^{-1} denotes the inverse complementary error function. This study systematically evaluates the generalization capability of neural networks by training them under different modulation formats. As shown in Fig. 6(a), the system performance in the linear region is predominantly governed by ASE noise, while in the nonlinear region it is mainly dominated by nonlinear effects, resulting in a characteristic inverted “U” shaped Q-factor curve. It is noteworthy that both the NMSE curve with ASE noise and the Q-factor curve reach their optimum at the same optimal transmission power, and this correspondence validates the accuracy of the proposed scheme.

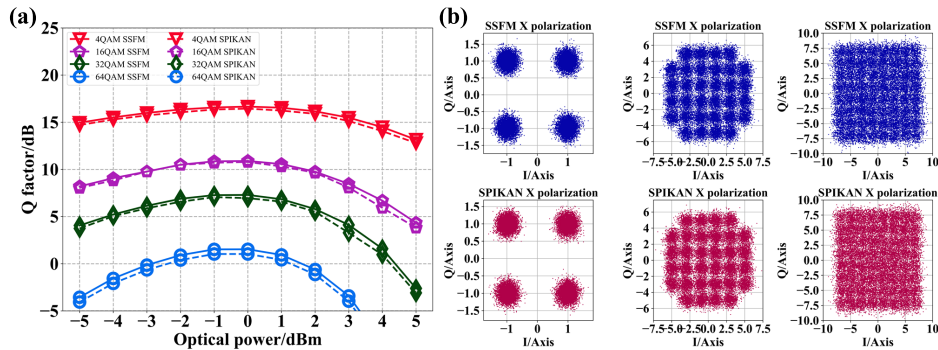


Fig. 6. (a) Q factors of different modulation formats at various transmitting powers. (b) Generalization of different modulation formats.

After being trained with 16QAM data transmitted over 15 spans, the SPIKAN model demonstrates excellent generalization capability across different modulation formats: for 4QAM modulation, its Q-factor differs by only 0.23 dB from the SSFM benchmark; under 32QAM modulation, the difference is 0.36 dB; and even with the highest-order 64QAM modulation, the Q-factor deviation remains merely 0.66 dB, as shown in Fig. 6(b). These findings indicate that

the proposed approach holds practical potential for optical fiber communication system design when the Q-factor tolerance in system design is within 0.66 dB.

4. Ultra-wideband transmission system

Subsequently, we extend and apply the proposed method to UWB WDM systems covering the $S + C + L$ bands. Compared to single-channel systems, signal transmission across these bands encounters more complex nonlinear effects, thereby significantly increasing the difficulty for neural networks to learn channel characteristics.

4.1. $S+C+L$ large band system

Owing to the substantial bandwidth characteristics of UWB systems, the conventional Manakov equation is no longer applicable, because the Manakov equation cannot accurately characterize the complex nonlinear dynamics inherent in UWB transmission [19]. Consequently, the coupled NLSE (CNLSE) is employed to describe the transmission characteristics of the system. The specific expression of the CNLSE is given as follows:

$$\begin{aligned} \frac{\partial E(z, t)}{\partial z} = & -\frac{\alpha}{2} E(z, t) + \frac{\Delta\beta_1}{2} \sigma \frac{\partial E(z, t)}{\partial t} + j \frac{\beta_2}{2} \frac{\partial^2 E(z, t)}{\partial t^2} \\ & - j \gamma \left[|E(z, t)|^2 E(z, t) - \frac{1}{3} \left(E^\dagger(z, t) \sigma_3 E(z, t) \right) \sigma_3 E(z, t) \right], \end{aligned} \quad (11)$$

where σ_3 represents the third component of the Pauli matrix, $E(z, t) = [E_x(z, t), E_y(z, t)]$, $E_x(z, t) = \sum_{i=1}^k E_x^i(z, t) \exp(j\Delta\omega_i t)$, $E_y(z, t) = \sum_{i=1}^k E_y^i(z, t) \exp(j\Delta\omega_i t)$, k is the total number of channels, $\Delta\omega_i = \omega_{ic} - \omega_0$ denotes the frequency difference between the carrier center frequency of the i -th channel ω_{ic} and the reference frequency ω_0 of the UWB signal. In the UWB transmission system covering $S + C + L$ bands, due to the extremely broad spectral range, the Kerr nonlinear coefficient must account for frequency dependency, expressed as $\gamma(\omega) = \omega n_2 / c A_{eff}$. By incorporating the effect of SRS and utilizing the SSFM described in Eq. (3), we have developed a simulation approach that accounts for SRS. This method has been validated in [19].

Based on the consideration of SRS in UWB systems, we have designed the loss function for SPIKAN, which is expressed as follows:

$$loss'_d = \frac{1}{N'_d} \sum_{i=1}^{N'_d} |\hat{s}'_i - s'_i|^2, loss'_p = \frac{1}{N'_p} \sum_{i=1}^{N'_p} (|f'_1|^2 + |f'_2|^2 + |f'_3|^2 + |f'_4|^2), \quad (12)$$

where f'_1 to f'_4 correspond to Eq. (26) to (27) in Appendix. Similarly to the single-carrier case, the neural network's output format matches the fiber terminal configuration, enabling direct computation of the data-driven loss component $loss'_d$. By substituting the derived expressions for f'_1, f'_2, f'_3 and f'_4 and the fiber parameters listed in Table 1 into Eq. (12), the physics-informed loss $loss'_p$ is calculated. This $loss'_p$ value is then substituted into Eq. (7) to determine the total loss function of the neural network.

As shown in Fig. 1(a), the system architecture in this study uses a reference wavelength of 1550 nm, with a channel spacing of 100 GHz and a total of 120 channels. Unlike single-channel scenarios, SRS imposes non-negligible dual detriments on system performance, arising from both nonlinear phase rotation and inter-channel power transfer, as shown in Fig. 7(a).

Figure 7(b) further delineates the power evolution of signals at different wavelengths along the fiber transmission distance in a UWB system with a reference wavelength of 1550 nm, 100 GHz channel spacing, and 120 channels total, illustrating the impact of SRS on system performance.

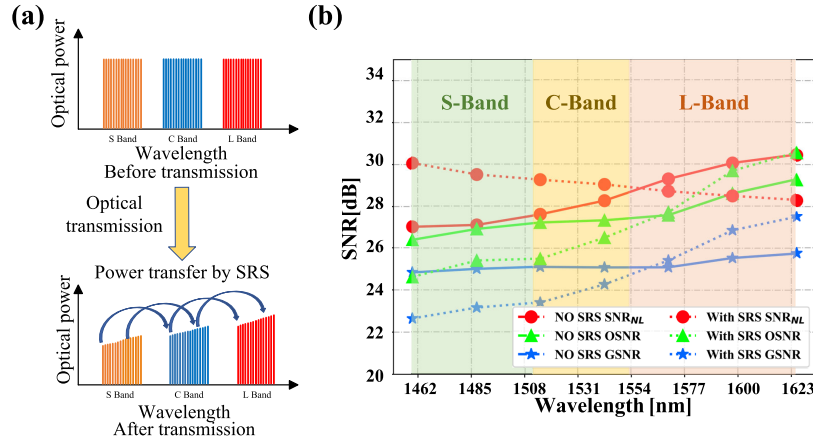


Fig. 7. (a) Power transfer effect of different wavelength signals in UWB systems induced by SRS, (b) Comparison of nonlinear SNR, OSNR, and GSNR in UWB systems with and without SRS consideration.

When SRS effects are not considered, it can be observed that the optical signal-to-noise ratio (OSNR) is lower than the nonlinear signal-to-noise ratio (SNR) [35], as the system is mainly influenced by linear effects, and nonlinear effects are negligible without SRS. However, when SRS are taken into account, the OSNR in the S-band decreases significantly, while the nonlinear SNR increases notably. In the L-band, the increase in OSNR due to SRS benefits the operation in the nonlinear region, compensating for the reduction in nonlinear SNR and resulting in an increased generalized signal-to-noise ratio (GSNR) [36]. The C-band serves as a transitional region between linear and nonlinear effects, with characteristics intermediate between the S-band and L-band. Figure 7(b) illustrates this phenomenon. To more accurately simulate actual UWB systems, all UWB data used in this study consider the impact of SRS.

4.2. Result

We first validated the modeling capability of the neural network, as shown in Fig. 8. To intuitively evaluate the modeling effectiveness of SPIKAN, we obtained the spectrum and time-domain plots at the fiber output end. Figure 8(a) to 8(f) show the spectral characteristics of the signal, where Fig. 8(a), 8(c), and 8(e) correspond to the S, C, and L bands, respectively. These spectral plots clearly demonstrate significant power transfer between different bands due to the presence of SRS effects. The curves of SPIKAN and SSFM exhibit nearly perfect overlap, demonstrating that SPIKAN successfully captures the impact of SRS effects on the power spectrum. Moreover, the temporal waveforms show high consistency, further validating the effectiveness and accuracy of the model. As shown in Fig. 8, the NMSE of the model is 3.06×10^{-3} , with a transmission distance of 800 km.

Subsequently, we validated the generalization capability of the three proposed networks under different transmit powers. As shown in Fig. 9, the NMSE of the neural networks gradually increases with the transmission distance, attributed to the cumulative effects of both linear and nonlinear phenomena as the distance increases. In Fig. 9, we employ data ranging from -4 dBm to 3 dBm for generalization using the model trained under 0 dBm condition. Even under the demanding conditions of 120 wavelength channels and an extended transmission distance of 800 km, SPIKAN maintains a low NMSE of 1.26×10^{-2} , demonstrating significantly better performance than the comparison models without physical constraints.

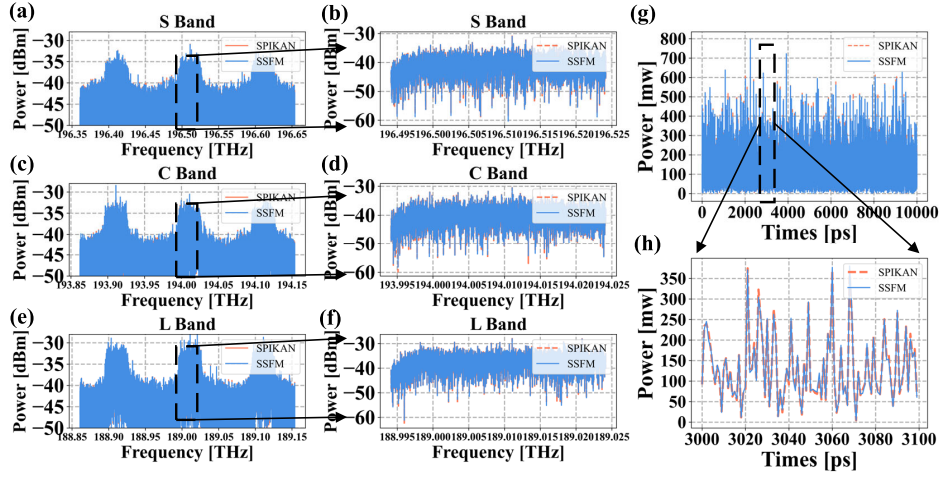


Fig. 8. (a) S-band spectrum, (b) Partial magnification of the S-band spectrum, (c) C-band spectrum, (d) Partial magnification of the C-band spectrum, (e) L-band spectrum, (f) Partial magnification of the L-band spectrum, (g) Time-domain waveform, (h) Partial magnification of the time-domain waveform.

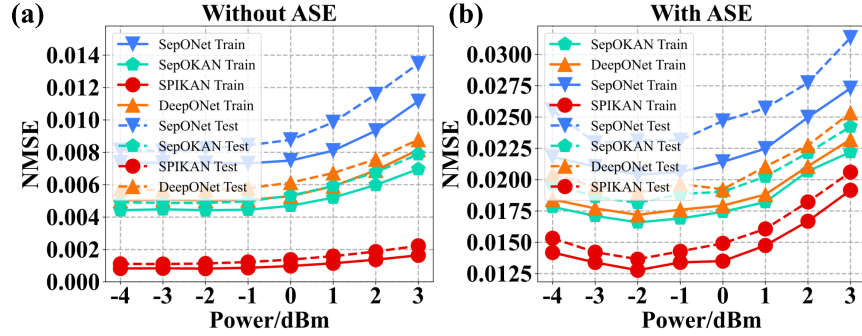


Fig. 9. Comparison of fitting performance in UWB Systems. (a) Without ASE, (b) With ASE.

4.3. Complexity analysis

We employ the number of real multiplications (RMs) as the metric for computational complexity, and conduct a systematic comparison between the proposed model, the conventional SSFM, and other neural network approaches. When using the SSFM to simulate fiber channel transmission, as shown in Eq. (2), two linear operators and one nonlinear operator are involved. Since the linear operators need to be computed in the frequency domain, the original signal must undergo a fast Fourier Transform (FFT). Since both the linear and nonlinear operators require frequency-domain computation, each involves $2N \log_2 N$ real multiplications. Given the step size L_{step} and the transmission span number L_{span} , we can derive the computational complexity expression per symbol for the SSFM method [18].

$$RM_{SSFM} = fL_{step}L_{span}(4N\log_2 N + rN)/N, \quad (13)$$

where f represents the upsampling factor, and r denotes the additional multiplication computations required beyond the FFT operations. As can be seen from Eq. (13), the computational complexity of the SSFM increases linearly with transmission distance. This implies that the computational

load grows proportionally with the transmission distance, thereby posing challenges for the real-time implementation of long-haul transmission systems.

When employing a MLP architecture for optical fiber channel modeling, we disregard the RMs required by nonlinear activation functions. Since a single layer of MLP can be represented as a matrix operation, with an input dimension of d_{in} and an output dimension of d_{out} , the required RMs amount to $d_{in} \times d_{out}$. We can derive the following expression for the RMs required per symbol by an MLP:

$$RMs_{MLP} = L_{span} \left(\frac{N - d_0/2f}{4} + 1 \right) \left(\sum_{i=0}^L d_i d_{i+1} \right) / N, \quad (14)$$

where d represents the number of neurons in one layer, L represents the total number of layers in a neural network, including the input layer, output layer, and hidden layers. When $i = 0$, it represents the dimensionality of the input layer; when $i = L$, it represents the dimensionality of the output layer.

When employing a KAN network for optical fiber channel modeling, due to $\phi(x) = w_b b(x) + w_s spline(x)$, the RMs of the KAN network primarily originate from two components [21]. First, we calculate the RMs of the first part, which for a network with an input of can be expressed as $d_{in} \times d_{out}$. The RMs of the second part are mainly caused by the spline function $spline(x)$, typically defined as $spline(x) = \sum_i c_i B_i(x)$. Given a spline order of k and a grid number of g , and disregarding the RMs induced by $B_i(x)$, the RMs of the second part can be obtained as $d_{in} \times d_{out} \times (k + g)$. Consequently, we can derive the following expression for the RMs required per symbol by a single KAN layer:

$$RMs_{KAN} = L_{span} \left(\frac{N - d_0/2f}{4} + 1 \right) \left(\sum_{i=0}^L d_i d_{i+1} \times (k + g) \right) / N. \quad (15)$$

As shown in Fig. 10, a comparison of the number of real multiplications per symbol for the five methods—SSFM, DeepONet, SepONet, SepOKAN, and SPIKAN—reveals the following findings: The results indicate that the computational complexity of SSFM increases linearly as the step size decreases, while DeepONet and SepONet exhibit moderate complexity. In contrast, the proposed SepOKAN and SPIKAN achieve the lowest complexity. Notably, SPIKAN reduces the number of real multiplications by nearly 70% compared to DeepONet, while maintaining comparable or even better accuracy. This convincingly demonstrates that the KAN architecture can achieve superior modeling performance while substantially reducing computational costs. To quantitatively compare modeling complexity and accuracy, we present Q-factor results for three SSFM step sizes alongside computational complexity metrics in Fig. 10. Furthermore, the parallel branch network design adopted in this work ensures that the overall complexity of multi-network modeling remains comparable to that of a single network, thereby effectively enhancing the overall computational efficiency. The neural network operates in an environment equipped with an i9-13900KF processor, a RTX 3060 Ti GPU, 64 GB of RAM, and utilizes Conda 4.12.0 and Python 3.8 software frameworks.

5. Experimental system results

Finally, the practical utility of the SPIKAN model was validated through real-world optical fiber transmission experiments, encompassing both single-channel and WDM system configurations.

5.1. Experimental system and data processing

This study constructed both single-channel and WDM experimental systems to validate the results presented in previous chapters. The experiments utilized G.652E fiber, with the system

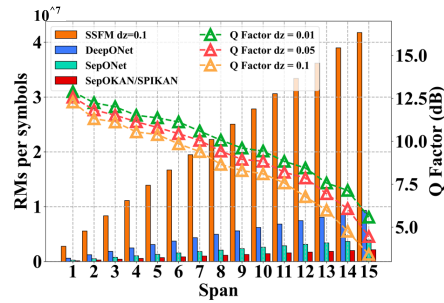


Fig. 10. Comparative analysis of computational complexity.

architecture illustrated in Fig. 11. Figure 11(a) shows the single-channel system architecture: at the transmitter end, IQ modulation is performed using an ECL with a linewidth of 100 kHz, and an AWG generates a 12 GBaud signal at a sampling rate of 25 GSa/s. This signal then undergoes processing through a polarization multiplexing module that includes a PM-OC, a PC, and a PBC. Finally, the optical signal is amplified by an EDFA and adjusted for transmit power using a VOA. After transmission through 10 spans of 80 km each, a coherent receiver containing a LO, a PBS, and a BPD splits the optical signal into four paths. These signals are then converted into electrical signals using a DPO with a sampling rate of 100 GSa/s and finally demodulated through DSP. Figure 11(b) illustrates the WDM system architecture: at the transmitter end, the WDM system constructed in this work comprises 61 channels, with target channel (channel of interest, COI) wavelength of 1550 nm. The 61 channels are multiplexed using a WSS and transmitted through 7 spans of 80 km each, the channel spacing is 100 GHz. At the receiver, an optical bandpass filter is employed to extract the COI from the WDM output signal for subsequent demodulation. This experimental system provides a reliable test platform for validating the proposed scheme in real-world conditions. Since the actual system cannot directly access optical signals transmitted through the fiber, we can only process the electrical signals after photoelectric conversion. To address this, we adopted the method described in Ref. [18] to eliminate additional interference caused by frequency offset and laser phase noise.

5.2. Single-channel system

The SPIKAN model, initially trained on simulated data, was fine-tuned using a portion of experimentally collected data and then evaluated on the remaining test dataset. As shown in Fig. 12, the output of the fine-tuned model (orange curve) aligns closely with the ground-truth link measurements. In contrast, the pre-fine-tuning model exhibits certain deviations, though these are substantially reduced after calibration with a limited amount of real-world data. The minimum NMSE achieved with experimental data is 1.52×10^{-2} , which is higher than that with simulated data. This discrepancy is mainly attributed to physical effects present in actual channels—such as stimulated Brillouin scattering—that were not considered in the simulations, as well as impairments introduced by the inherent characteristics of practical devices. Nevertheless, the time-domain waveform comparison in Fig. 12(b) demonstrates that the SPIKAN output remains highly consistent with the actual link data, validating the efficacy of the proposed method in real-world applications.

This study systematically evaluates the performance of the SPIKAN model by testing signals with power levels ranging from -5 dBm to 5 dBm, as shown in Fig. 13. The optimal signal quality is achieved when the input power reaches a critical point where linear and nonlinear effects are balanced. This trend is consistent with the results obtained from pure numerical simulations. Specifically, the differences in the Q-factor predicted by the SPIKAN model compared to the

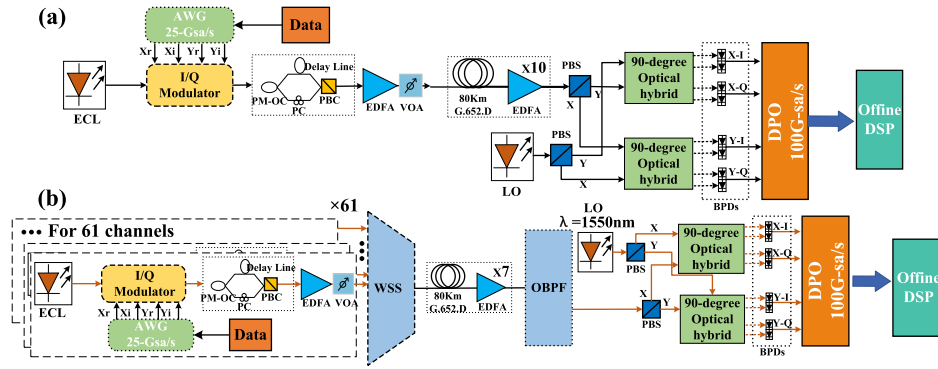


Fig. 11. The experimental link systems. (a) Single-channel dual-polarization system architecture; (b) WDM system architecture. The abbreviations in the figure represent: AWG: arbitrary waveform generator; ECL: external cavity laser; PDM: polarization division multiplex; PM-OC: polarization-maintaining optical coupler; PC: polarization controller; PBC: polarization beam combiner; EDFA: erbium-doped fiber amplifier; VOA: variable optical attenuator; SSF: standard single-mode fiber; LO: local oscillator; PBS: polarization beam splitter; BPD: balanced photonic detector; DPO: digital phosphor oscilloscope; OBPF: optical bandpass filter; WSS: wavelength-selective switch.

actual measured values at input powers of -5 dBm, 0 dBm, and 5 dBm are 0.36 dB, 0.20 dB, and 0.54 dB, respectively. It is noteworthy that the difference in the Q-factor corresponding to the optimal transmission power predicted by the SPIKAN model and that of the actual link is merely 1.3%. The minimal error fully demonstrates the exceptional accuracy of the SPIKAN model in practical optical fiber links.

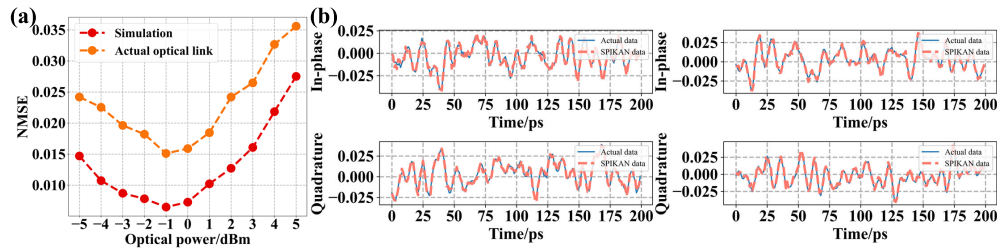


Fig. 12. (a) Comparative analysis of NMSE between experimental and simulated links, (b) Comparative analysis of time-domain between experimental and simulated links.

5.3. WDM system

The experiment tests the NMSE of a single-channel carrying a signal over a range of -4 dBm to 2 dBm across 7 spans, as shown in Fig. 14(a). The red curve represents the NMSE values obtained from simulation data, while the orange curve represents the NMSE values obtained from actual link data. The experimental results indicate that due to the more complex influencing factors in WDM systems compared to single-channel systems, SPIKAN faces greater challenges in modeling WDM systems. Various influencing factors present in actual links make the modeling results based on actual link data more complex than those based on simulation data. However, even under the condition of transmitting over 7 spans, SPIKAN achieves an NMSE performance of approximately 1.81×10^{-2} when applied to actual link data. To visually demonstrate the waveform at this NMSE level, we compare the output of the SPIKAN method with the actual

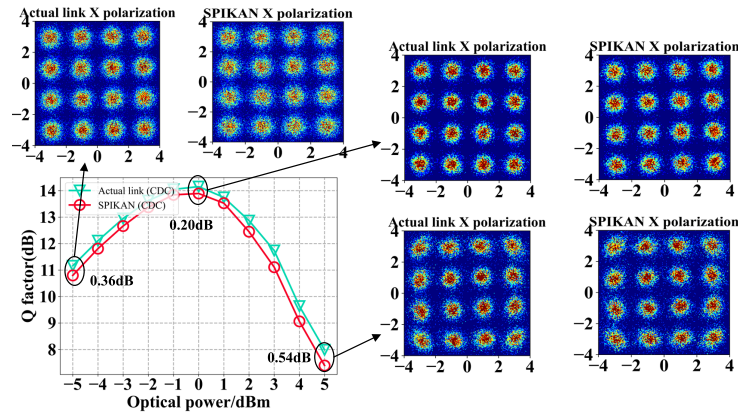


Fig. 13. Comparative analysis of demodulated data.

link data, as shown in Fig. 14(b). Visually, the waveforms are largely consistent, validating the effectiveness of the SPIKAN method. The comprehensive experimental results demonstrate that the SPIKAN model can accurately predict link performance, indicating its strong practical utility for engineering applications.

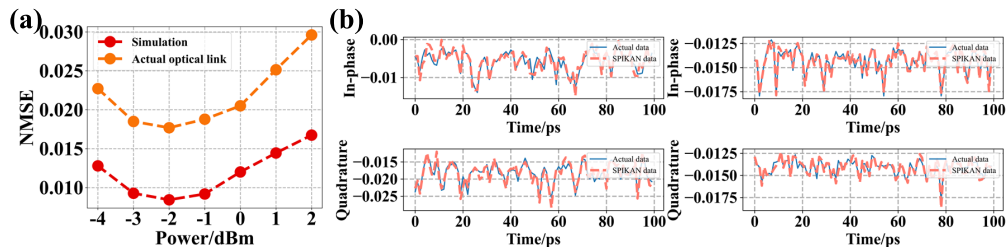


Fig. 14. (a) Comparative analysis of NMSE between experimental and simulated links in WDM systems, (b) Comparative analysis of time-domain between experimental and simulated links in WDM systems.

6. Conclusion

This study systematically constructs three optical fiber channel modeling schemes by effectively integrating the architectural concepts of SepONet, KAN, and physics-informed neural networks: first, an operator learning model based on the SepONet framework; subsequently, the SepOKAN model formed by incorporating the KAN architecture; and finally, the SPIKAN model with physical interpretability developed through the integration of physical constraints. In simulation verification, SPIKAN achieves an NMSE of 7.77×10^{-3} after 1200 km single-channel transmission and an NMSE of 1.26×10^{-2} after 800 km UWB transmission with ASE, both results being significantly better than those of comparative models. In practical experiments, SPIKAN exhibits merely a 0.20 dB Q-factor difference from measured values with an NMSE of 1.52×10^{-2} over an 800 km single-channel actual link, and with an NMSE of 1.81×10^{-2} over a 560 km WDM link. The results demonstrate that SPIKAN exhibits outstanding power adaptability and superior modulation format generalization capability. Furthermore, its remarkable computational efficiency makes it a compelling candidate for optical fiber communication system simulation, indicating substantial application potential.

Appendix. Formula derivation details

A.1. Single-channel system

Assuming $E(z, t) = [E_x(z, t), E_y(z, t)]$, $E_x(z, t) = E_x^r(z, t) + jE_x^i(z, t)$, $E_y(z, t) = E_y^r(z, t) + jE_y^i(z, t)$ in Eq. (1), Eq. (1) can be rewritten as the following four decoupled equations:

$$D_x^r = -P_s E_x^i(z, t), D_x^i = P_s E_x^r(z, t), D_y^r = -P_s E_y^i(z, t), D_y^i = P_s E_y^r(z, t), \quad (16)$$

$$P_s = \gamma \frac{8}{9} (|E_x^r(z, t)|^2 + |E_x^i(z, t)|^2 + |E_y^r(z, t)|^2 + |E_y^i(z, t)|^2), \quad (17)$$

where E_x^r , E_x^i , E_y^r , and E_y^i represent the real and imaginary parts of the X-polarization and the real and imaginary parts of the Y-polarization of the optical signal, respectively. In addition, to save space and simplify calculations, we define the linear parts of the equations as D_x^r , D_x^i , D_y^r , D_y^i :

$$D_x^r = \frac{\alpha}{2} E_x^r + \frac{\partial E_x^r}{\partial z} - \frac{\beta_2}{2} \frac{\partial^2 E_x^r}{\partial t^2} - \frac{\Delta\beta_1}{2} \left[\sigma_{xx} \frac{\partial E_x^r}{\partial t} + \sigma_{xy} \frac{\partial E_y^r}{\partial t} \right], \quad (18)$$

$$D_x^i = \frac{\alpha}{2} E_x^i + \frac{\partial E_x^i}{\partial z} - \frac{\beta_2}{2} \frac{\partial^2 E_x^i}{\partial t^2} - \frac{\Delta\beta_1}{2} \left[\sigma_{xx} \frac{\partial E_x^i}{\partial t} + \sigma_{xy} \frac{\partial E_y^i}{\partial t} \right], \quad (19)$$

$$D_y^r = \frac{\alpha}{2} E_y^r + \frac{\partial E_y^r}{\partial z} - \frac{\beta_2}{2} \frac{\partial^2 E_y^r}{\partial t^2} - \frac{\Delta\beta_1}{2} \left[\sigma_{yx} \frac{\partial E_x^r}{\partial t} + \sigma_{yy} \frac{\partial E_y^r}{\partial t} \right], \quad (20)$$

$$D_y^i = \frac{\alpha}{2} E_y^i + \frac{\partial E_y^i}{\partial z} - \frac{\beta_2}{2} \frac{\partial^2 E_y^i}{\partial t^2} - \frac{\Delta\beta_1}{2} \left[\sigma_{yx} \frac{\partial E_x^i}{\partial t} + \sigma_{yy} \frac{\partial E_y^i}{\partial t} \right]. \quad (21)$$

Based on this, four formulas for the physical constraints in single-channel systems can be constructed:

$$f_1(z, t) = D_x^r + P_s E_x^i(z, t), f_2(z, t) = D_x^i - P_s E_x^r(z, t), \quad (22)$$

$$f_3(z, t) = D_y^r + P_s E_y^i(z, t), f_4(z, t) = D_y^i - P_s E_y^r(z, t). \quad (23)$$

A.2. WDM system

Eq. (11) can likewise be expressed as four decoupled equations:

$$D_x^r = \gamma(1 - f_r)A_1 + \gamma f_r B_1, D_x^i = -\gamma(1 - f_r)A_2 - \gamma f_r B_2, \quad (24)$$

$$D_y^r = \gamma(1 - f_r)A_3 + \gamma f_r B_3, D_y^i = -\gamma(1 - f_r)A_4 - \gamma f_r B_4, \quad (25)$$

where $A_1 = 2R_x E_x^r E_x^i + \frac{2}{3}|E_x|^2 E_y^r E_y^i$, $B_1 = 2H_x E_x^r E_x^i + H(E_x^i E_y^r + E_x^r E_y^i)$, $A_2 = R_x C_x + \frac{1}{3}|E_x|^2 C_y$, $B_2 = H_x C_x + H(E_x^i E_y^r - E_x^r E_y^i)$, $A_3 = 2R_y E_y^r E_y^i + \frac{2}{3}|E_y|^2 E_x^r E_x^i$, $B_3 = 2H_y E_y^r E_y^i + H(E_x^i E_y^r + E_x^r E_y^i)$, $A_4 = R_y C_y + \frac{1}{3}|E_y|^2 C_x$, $B_4 = H_y C_y + H(E_x^i E_y^r - E_x^r E_y^i)$, $R_x = |E_x|^2 + \frac{2}{3}|E_y|^2$, $R_y = |E_y|^2 + \frac{2}{3}|E_x|^2$, $C_x = (E_x^r)^2 - (E_x^i)^2$, $C_y = (E_y^r)^2 - (E_y^i)^2$, $H_x = h_1 * |E_x|^2 + h_2 * |E_y|^2$, $H_y = h_1 * |E_y|^2 + h_2 * |E_x|^2$, $H = h_3 * (2E_x^r E_y^r + 2E_x^i E_y^i)$, h_1 , h_2 and h_3 and represent the defined Raman response functions, respectively. f_r is the Raman fraction. The introduction of these parameters enables the loss function to accurately capture the polarization-dependent power transfer characteristics in UWB systems caused by SRS. Based on this, four formulas for the physical constraints in UWB systems can be constructed:

$$f_1'(z, t) = D_x^r - \gamma(1 - f_r)A_1 - \gamma f_r B_1, f_2'(z, t) = D_x^i + \gamma(1 - f_r)A_2 + \gamma f_r B_2, \quad (26)$$

$$f_3'(z, t) = D_y^r - \gamma(1 - f_r)A_3 - \gamma f_r B_3, f_4'(z, t) = D_y^i + \gamma(1 - f_r)A_4 + \gamma f_r B_4. \quad (27)$$

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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